

A Six-Parameter Extension of the Direct Somos–4 Algebraic Generating Function

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Abstract

The preceding paper constructs a five-parameter quadratic algebraic generating function with prescribed initial Hankel determinants (λ, m, r, η) and Somos–4 parameter τ . We adjoin a second Somos coefficient α . If $q^6 = \alpha$, the weighted transport

$$G_{\alpha; \lambda, m, r, \eta, \tau}(x) = G_{1; \lambda, q^{-2}m, q^{-6}r, q^{-12}\eta, q^{-8}\tau}(qx)$$

preserves the four prescribed ordinary Hankel determinants and changes the recurrence to

$$H_{k+4}H_k = \alpha H_{k+3}H_{k+1} + \tau H_{k+2}^2.$$

The once-shifted determinants acquire an explicit grading factor. We also transport the discriminant exactly and show that its residual genus-one quartic has the rational model

$$Y^2 = (1 - (A + \alpha B)z + \tau z^2)^2 - 4\alpha^2 z^3,$$

where A and B are rational functions of the initial Hankel data. This separates curve-level descent from descent of the distinguished formal-power-series branch. The latter remains open; no nonexistence claim is made.

1 Introduction

The five-parameter construction in [1] uses the notation

$$(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta)$$

and the recurrence parameter τ in

$$H_{k+4}H_k = H_{k+3}H_{k+1} + \tau H_{k+2}^2, \quad k \geq 1. \tag{1}$$

We retain that notation throughout. The new parameter α is the coefficient of the first bilinear term, so the six-parameter recurrence is

$$H_{k+4}H_k = \alpha H_{k+3}H_{k+1} + \tau H_{k+2}^2, \quad k \geq 1. \tag{2}$$

Thus the parameter list is $(\lambda, m, r, \eta, \alpha, \tau)$, and specialization $\alpha = 1$ recovers the normalization of [1].

The extension is obtained by a weighted scaling over a field containing a sixth root of α . The determinant identities and the transported discriminant are exact formal identities. They do not require expansion of the lengthy quadratic polynomial from the preceding paper. What is not settled by these identities is whether the distinguished normalized series itself descends from the sixth-root extension to $\mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$.

2 The five-parameter input

Let

$$G_{1;\lambda,m,r,\eta,\tau}(x) = \sum_{k \geq 0} g_k x^k$$

be the selected algebraic series of [1]. Define

$$H_N(g) = \det(g_{i+j})_{0 \leq i,j \leq N-1}, \quad D_N(g) = \det(g_{i+j+1})_{0 \leq i,j \leq N-1}.$$

The normalization in the preceding paper is

$$(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta), \quad D_N = H_{N+2}. \quad (3)$$

Its quadratic discriminant factors as

$$\Delta_1(x) = (x\lambda^2\eta mr^2)^2 \left((1 - Jx + \tau x^2)^2 - 4x^3 \right), \quad (4)$$

where

$$J = \frac{\lambda\eta}{mr} + \frac{m^2}{\lambda r} + \frac{r^2}{m\eta} + \frac{\tau mr}{\lambda\eta}. \quad (5)$$

These statements are the only inputs from the five-parameter construction needed below.

3 Weighted transport

Work over

$$\mathbb{K} = \mathbb{Q}(\lambda, m, r, \eta, q, \tau), \quad q^6 = \alpha,$$

and introduce the primed parameters

$$\lambda' = \lambda, \quad m' = q^{-2}m, \quad r' = q^{-6}r, \quad \eta' = q^{-12}\eta, \quad \tau' = q^{-8}\tau. \quad (6)$$

Define

$$G_{\alpha;\lambda,m,r,\eta,\tau}(x) := G_{1;\lambda',m',r',\eta',\tau'}(qx). \quad (7)$$

If

$$G_{1;\lambda',m',r',\eta',\tau'}(x) = \sum_{k \geq 0} a_k x^k, \quad G_{\alpha;\lambda,m,r,\eta,\tau}(x) = \sum_{k \geq 0} b_k x^k,$$

then

$$b_k = q^k a_k. \quad (8)$$

Proposition 1 (Determinant weights). *For every $N \geq 1$,*

$$H_N(b) = q^{N(N-1)} H_N(a), \quad D_N(b) = q^{N^2} D_N(a). \quad (9)$$

Proof. In the ordinary Hankel matrix, its (i, j) entry is multiplied by q^{i+j} . Factoring q^i from row i and q^j from column j gives the exponent $2 \sum_{j=0}^{N-1} j = N(N-1)$. For the shifted matrix the entry is multiplied by q^{i+j+1} . The additional factor q in every row contributes N , so the total exponent is $N(N-1) + N = N^2$. $\square$

4 The two-coefficient Somos–4 recurrence

Theorem 2. *Let $\widehat{H}_N = H_N(b)$. Then*

$$(\widehat{H}_1, \widehat{H}_2, \widehat{H}_3, \widehat{H}_4) = (\lambda, m, r, \eta)$$

and

$$\widehat{H}_{k+4}\widehat{H}_k = \alpha\widehat{H}_{k+3}\widehat{H}_{k+1} + \tau\widehat{H}_{k+2}^2, \quad k \geq 1. \quad (10)$$

Proof. Equation (9) and (6) give

$$\widehat{H}_1 = \lambda, \quad \widehat{H}_2 = q^2 m' = m, \quad \widehat{H}_3 = q^6 r' = r, \quad \widehat{H}_4 = q^{12} \eta' = \eta.$$

Write $H'_N = H_N(a)$. The primed sequence satisfies

$$H'_{k+4}H'_k = H'_{k+3}H'_{k+1} + \tau'H_{k+2}^2.$$

The weight difference between the left-hand product and the first product on the right is 6, while that between the left-hand product and H_{k+2}^2 is 8. Hence

$$\widehat{H}_{k+4}\widehat{H}_k = q^6\widehat{H}_{k+3}\widehat{H}_{k+1} + q^8\tau'\widehat{H}_{k+2}^2.$$

Now use $q^6 = \alpha$ and $q^8\tau' = \tau$. □

Corollary 3 (Low-order recurrence check). *The first two values forced beyond the initial tuple are*

$$\widehat{H}_5 = \frac{\alpha\eta m + \tau r^2}{\lambda}, \quad \widehat{H}_6 = \frac{\alpha\widehat{H}_5 r + \tau\eta^2}{m}.$$

These formulas provide a direct indexing check for any coefficient expansion of (7).

5 The transported companion normalization

Proposition 4. *If the primed series obeys $D_N(a) = H'_{N+2}$, then*

$$\widehat{D}_N := D_N(b) = q^{-3N-2}\widehat{H}_{N+2}, \quad N \geq 1. \quad (11)$$

Proof. By (9),

$$D_N(b) = q^{N^2}H'_{N+2}, \quad \widehat{H}_{N+2} = q^{(N+2)(N+1)}H'_{N+2}.$$

Their exponent difference is $N^2 - (N+2)(N+1) = -3N-2$. □

Remark 5. A constant coefficient rescaling $b_k \mapsto cb_k$ changes the ratio $D_N(b)/\widehat{H}_{N+2}$ by c^{-2} . It cannot remove the factor q^{-3N} . Thus the literal identity $D_N = H_{N+2}$ is not invariant under this graded transport. Equation (11), rather than the ungraded identity, is the companion normalization naturally inherited from the five-parameter series.

6 Exact transport of the discriminant

Separate the old invariant into

$$A = \frac{\lambda\eta}{mr} + \frac{\tau mr}{\lambda\eta}, \quad B = \frac{m^2}{\lambda r} + \frac{r^2}{m\eta}. \quad (12)$$

After substitution of (6),

$$J' = q^{-4}A + q^2B. \quad (13)$$

Consequently the substitution $x \mapsto qx$ gives

$$qJ' = q^{-3}A + q^3B = \alpha^{-1/2}A + \alpha^{1/2}B. \quad (14)$$

Proposition 6 (Transported discriminant). *The discriminant obtained by substituting (6) and $x \mapsto qx$ into (4) is*

$$\Delta_\alpha(x) = q^{-50}(x\lambda^2\eta mr^2)^2 \left[\left(1 - (q^{-3}A + q^3B)x + q^{-6}\tau x^2 \right)^2 - 4q^3x^3 \right]. \quad (15)$$

Equivalently,

$$\Delta_\alpha(x) = \alpha^{-25/3}(x\lambda^2\eta mr^2)^2 \left[\left(1 - (\alpha^{-1/2}A + \alpha^{1/2}B)x + \frac{\tau}{\alpha}x^2 \right)^2 - 4\alpha^{1/2}x^3 \right]. \quad (16)$$

Proof. The square prefactor has total q -weight $2(1 - 12 - 2 - 12) = -50$. The three terms in the residual factor transform as $J'x \mapsto qJ'x$, $\tau'x^2 \mapsto q^2\tau'x^2 = q^{-6}\tau x^2$, and $x^3 \mapsto q^3x^3$. Equations (13) and $q^6 = \alpha$ give the stated forms. $\square$

The overall monomial factor in (15) is a square in $\mathbb{K}(x)$ and does not affect the residual double cover. It should, however, be retained when comparing discriminants of particular polynomial representatives, because multiplying a quadratic equation by a scalar squares that scalar in its discriminant.

7 A rational curve-level model

Remove the square prefactor in (16) and put

$$x = \alpha^{1/2}z. \quad (17)$$

The residual equation becomes

$$Y^2 = \left(1 - (A + \alpha B)z + \tau z^2 \right)^2 - 4\alpha^2 z^3. \quad (18)$$

All coefficients of (18) belong to $\mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$.

For comparison with the quartic convention used in [1], set $X = z^{-1}$ and $Y = z^{-2}y$. Then (18) is birational to

$$y^2 = \left(X^2 - (A + \alpha B)X + \tau \right)^2 - 4\alpha^2 X. \quad (19)$$

Thus the curve descends to the rational six-parameter field. The coordinate change (17) nevertheless rescales coefficients of a formal power series and does not preserve the normalized Hankel data. Curve descent and descent of the distinguished generating series are therefore different problems.

8 What remains unresolved

The transport (7) defines a distinguished algebraic series over $\mathbb{K}$. The preceding results prove the ordinary Hankel recurrence, the graded companion identity, and the discriminant factorization. They do not prove that the coefficients of this distinguished branch belong to

$$\mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau).$$

Nor do they prove that no Hankel-compatible rationalizing transformation exists. Such a statement would require both a precisely specified class of admissible transformations and an invariant obstructing descent.

A complete series-level treatment should therefore supply the explicit transported quadratic equation, remove its common Laurent monomial and polynomial content, select the branch by its constant term, and determine the Galois action $q \mapsto \zeta q$ with $\zeta^6 = 1$ on that branch. These are open tasks rather than assumptions used in the results above.

9 Conclusion

The weighted substitution (7) gives an exact six-parameter extension of the direct Somos–4 construction. In notation consistent with [1], the parameters are $(\lambda, m, r, \eta, \alpha, \tau)$. The prescribed initial Hankel tuple is unchanged, the recurrence becomes (10), and the companion determinants satisfy (11). Exact transport of the five-parameter discriminant gives (15); after a coordinate rescaling, its residual genus-one curve has the rational model (18). The remaining issue is descent of the selected normalized formal series, not descent of the curve.

Disclosure on authoring and verification tools

The manuscript was revised with assistance from AI-based language and symbolic calculation tools. Their use was directed toward notation control, LaTeX editing, and exact algebraic checks. The author remains responsible for the mathematical statements and the final text.

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